

ABA-VR: Training Applied Behaviour Analysis Practitioners in Virtual Reality to Support Autistic Children

Scott Naylor*
Aalborg University

Ali Adjorlu†
Aalborg University



Figure 1: A virtual environment for contextual Applied Behavioural Analysis. Screenshot from the ABA-VR application.

ABSTRACT

Teachers of students with Autism Spectrum Disorder (ASD) often face significant challenges due to limited access to specialised training in Applied Behavioural Analysis (ABA). This position paper proposes a Virtual Reality (VR) application to provide accessible, contextual, and affordable ABA training for educators. By simulating realistic scenarios, such as classroom management or social interactions, the VR platform enables experienced ABA professionals to train educators through role-play and personalised feedback. The use of networked multiplayer functionality ensures that training transcends socioeconomic and geographical boundaries, democratising access to high-quality ABA instruction. This paper discusses the design of the VR platform and its potential to enhance educational outcomes for both teachers and students with ASD.

Index Terms: Autism, ASD, Applied Behavioural Analysis, ABA, Virtual Reality, VR.

1 INTRODUCTION

Autism Spectrum Disorder (ASD) is a prevalent neurodevelopmental condition. The Autism and Developmental Disabilities Monitoring (ADDM) has been tracking ASD rates among 8 year old children in the United States since the year 2000, where 67 per 10,000

were diagnosed [11], in 2020 the rates were 276 per 10,000 [9], indicating a 312% increase. ASD prevalence varies from nation to nation, and the cause is a complicated and currently debated topic. It could arise from increased awareness, methodologies of data collection, differences in diagnostic criteria, cultural perceptions, or environmental factors [14].

Regardless of the cause of increasing prevalence, ensuring that students with ASD are supported in our educational systems could not only improve the quality of life of the individual but also foster a more inclusive and productive society. Students with ASD often face significant behavioural and communication challenges, including self-injurious and aggressive behaviours. A 2021 survey involving 2,327 children with ASD found that over 40% exhibited self-injurious behaviours, aggressive behaviours, or both [4]. These challenges can have a catastrophic impact on a child's ability to thrive in social and educational settings, while also negatively affecting the quality of education for their peers. One approach to addressing these issues is to supply special education teachers with innovative tools to better understand and interact with students with ASD. Such tools could enable teachers to practice managing challenging situations in ways that benefit both the autistic student and their peers, ultimately improving educational outcomes for all.

Research into Applied Behavioural Analysis (ABA) has shown promising results as a framework for empowering special education teachers to deal with situations that may arise when working with students with ASD [6]. ABA utilises systematic techniques that allow the practitioner to create individualised interventions for the student. This involves analysing problematic behaviours, isolating them and implementing strategies to manage those behaviours [3]. However, some studies cite the failure to comply with ABA

*e-mail: sjna@create.aau.dk

†e-mail: adj@create.aau.dk

techniques as a reason for the inefficacy of ABA in assistive technologies [1]. This could be remedied by creating a platform that allows existing professionals in ABA to train new teachers in the philosophies and techniques of ABA.

This position paper proposes the development of a Virtual Reality (VR) application as part of an online training platform to make ABA training more accessible and affordable. Compared to physical training, VR headsets are relatively inexpensive, this facilitates a wider reach across diverse educational settings. Using VR, educators experienced in ABA could simulate common "problem" scenarios and role-play as students with ASD. This enables experienced educators to train new teachers in ABA techniques through meaningful and contextual environments. This approach could help to democratise access to relevant and important training and through a networked multiplayer approach: reach a diverse user base, regardless of location or economic boundaries.

2 THE PROBLEM

2.1 Current state of teacher training for ASD

A 2020 study using an online questionnaire with 78 primary school teachers found that a majority of teachers held neutral or negative views regarding the inclusion of children with ASD. Interestingly, factors such as years of teaching experience or training within special education teaching did not significantly influence their responses [8]. From this research, one could ask: is the efficacy of special education teacher training being adequately prioritised in our educational systems considering the significant increase in ASD diagnoses?

It is clear that the quality and access to teacher training for ASD will naturally vary across socioeconomic boundaries. However, we believe access to a high standard of special education training should not be limited to those who were lucky enough to be born in an affluent environment. We also believe that this training would not only bolster teachers' ability to support students with ASD but would also positively impact the lives of everyone involved: students, teachers, and their families.

2.2 School Non-Attendance and its Implications for Students with ASD

Another study from 2020 examined school non-attendance among children with ASD. This study surveyed 486 parents of children with ASD (average age 11 years), analysing school attendance over a period of one month. The findings revealed that, on average, students with ASD missed five out of 23 school days during the month. Furthermore, 43% of the students missed 10% or more of available school days [15].

The reasons for school non-attendance were diverse:

- "School refusal" accounted for 43% of absences and was the most frequently cited reason.
- Non-problematic absenteeism, usually medical appointments or illness related absences, accounted for 32%.
- School exclusion and school withdrawal each represented 9% of absences.
- Instances of truancy were almost negligible.

The study concludes the level of persistent non-attendance to be concerning and argues for the prioritisation of more support for students with high levels of persistent absence.

2.3 The Broader Implications of School Non-Attendance of students with ASD

We have presented research that shows a correlation between ASD and non-attendance in school, missing these vital early stage interaction and mental growth opportunities has shown to have catastrophic impacts on students across the neurological spectrum. Absenteeism has been intrinsically linked to physical conditions such as asthma and, psychologically, anxiety, depression and disruptive behaviour disorders [7].

Furthermore, a study on students with ASD reintegrating into school life following an exclusion found the experiences of the parents and children were overwhelmingly negative [2]. This included inadequate approaches to the students' behaviour and a general lack of understanding which lead to both the initial exclusion and hardship in integration. This study identified four key factors for successful reintegration:

- Substantial adjustments to the physical environment.
- Promoting strong staff-student relationships
- Understanding students' specific needs
- Targeted efforts towards improving students' wellbeing.

These factors align with the ABA approach of handling situations that may arise when a teacher is faced with a problematic scenario involving a student with ASD and highlights a requirement for a novel approach to teacher training regarding ASD.

3 RELATED RESEARCH

3.1 VRTeacher

VRTeacher is an ERASMUS+ funded initiative that has similar goals to ABA-VR. In VRTeacher, teachers are trained to handle crisis scenarios in contextual VR environments. The studies argue that VR-based teacher training can offer a credible alternative to in-person training and facilitates experimentation while negating any risk of actual harm to a student. VRTeacher found that teachers gained empathy and a better understanding of the needs of the students [12] [13].

VRTeacher offers a novel approach to teacher training through the use of virtual reality. However, the scenarios within the experience are fixed and predefined. Networking such a tool could enable the training to be conducted in real-time, remotely, from anywhere in the world. This could also improve access to ABA professionals and increase the variety of scenarios through role-play.

3.2 ABA Approach to Autism

ABA is an evidence-based intervention widely used to support individuals with ASD. Rooted in behavioural science, ABA focuses on teaching new skills, reducing maladaptive behaviours, and improving social and functional outcomes. This framework emphasises structured, individualised interventions tailored to the specific needs of each individual, making it highly versatile for diverse populations [3].

Research demonstrates the effectiveness of ABA in achieving significant developmental gains for children with ASD, particularly when applied early in life. Early intensive behavioural interventions grounded in ABA principles have been shown to enhance cognitive, social, and language skills, often leading to long-term improvements in quality of life [5]. These programs have become a cornerstone of evidence-based practice for ASD [10], reflecting their empirical support across a range of studies.

Despite its efficacy, ABA's success relies heavily on proper implementation. Inconsistent adherence to ABA principles, often due to inadequate training, can reduce its effectiveness [10]. This highlights the need for innovative solutions to bridge gaps in training

accessibility. Virtual reality provides a promising avenue for addressing this challenge, enabling immersive, context-rich environments where educators and caregivers can practice ABA techniques in realistic scenarios.

By leveraging VR, educators can gain hands-on experience with ABA strategies, democratising access to high-quality training and improving outcomes for individuals with ASD globally.

4 OUR PROPOSED SOLUTION

The proposed VR intervention is a training platform to help individuals learning Applied Behavioural Analysis (ABA) practice ABA techniques in a virtual environment. The platform allows an ABA teacher and an ABA student to join the same VR environment, where they can collaboratively engage in role-playing exercises. The ABA trainee practices applying ABA strategies, while the ABA teacher controls a child's avatar and provides realistic behavioural responses and real-time feedback. This setup creates a safe and controlled environment for trainees to refine their skills, bridging the gap between theoretical learning and practical application. The application is developed using the Unity3D game engine¹, with networking implemented via Photon Fusion², and 3D models either created in Blender 3D³ or sourced from BlenderKit⁴.

The environments and interactions were designed collaboratively during a workshop involving ABA professionals, where novel VR demos were presented to the participants, followed by a brainstorming session to devise the requirements of the solution.

For the remainder of this paper, the authors will refer to the ABA trainee (role playing as the teacher) as the "*teacher*" and the ABA expert (role playing as the child with ASD) as the "*student*".

4.1 Avatars

It is important, due to the personalised nature of ABA that the VR experience is capable of personalisation. This was done through the use of avatar customisation. For the teacher, two base adult avatars can be chosen from and subsequently customised through parameters such as skin colour, hairstyle, hair colour and some outfit options. The child has similar customisation options but with two base child avatar options, see figure (2).



Figure 2: Base choice of avatars for the teacher and student. Renderings of avatars from the ABA-VR application.

4.2 Environments and Scenarios

The VR experience begins in a bedroom environment, this represents the safe space of the child. An optional tutorial introduces the user to the various controls and interactions before they choose their base avatar and begin the personalisation process.

The main experience includes three contextual environments—classroom, supermarket, and home—designed to immerse users in realistic settings where challenging behaviours may occur. These environments are crafted to provide trainees with opportunities to practice applying ABA techniques in meaningful contexts. Each environment contains interactive props and customisable elements to support role-playing and simulate real-life scenarios.

The classroom environment (see figure: 3) replicates a typical school setting, with desks, chairs, and a whiteboard. The supermarket environment (see figure: 5) mimics a shopping aisle with shelves stocked with items and a checkout area. Finally, the home environment (see figure: 4) includes a living room, dining table, and other everyday objects, offering a familiar space for simulating domestic scenarios.

Within these environments, the following scenarios are designed to provide relevant training opportunities:

In the classroom environment, one scenario involves a student repeatedly calling out answers without raising their hand, disrupting the lesson. The teacher must apply ABA strategies to redirect the behaviour, such as prompting the student to follow classroom rules and reinforcing appropriate participation. Another scenario could involve a student refusing to engage in group activities, requiring the teacher to encourage inclusion through positive reinforcement and clear communication.

In the supermarket environment, trainees may encounter a scenario where a child becomes upset after being denied a favorite snack, escalating into a tantrum. The caregiver, role-played by the teacher, must use distraction or redirection techniques to de-escalate the situation. Another scenario could involve the child wandering away from the caregiver, requiring immediate action to ensure the child's safety while managing their distress.

In the home environment, a scenario might involve the teacher needing to make an important phone call while a child demands immediate attention. The teacher must manage the child's behaviour

¹<https://unity.com/>

²<https://www.photonengine.com/Fusion>

³<https://www.blender.org/>

⁴<https://www.blenderkit.com/>

using ABA strategies, such as setting clear expectations, offering alternative activities, or reinforcing patience. Another potential scenario could involve transitioning the child from screen time to a structured task, emphasising the use of reinforcement and clear communication.

To ensure the effectiveness and relevance of these scenarios, we plan to collaborate closely with ABA professionals during their design and refinement. This partnership will help ensure that the platform provides practical and meaningful training for educators in a variety of challenging situations.



Figure 3: Classroom environment. Screenshot from the ABA-VR application.

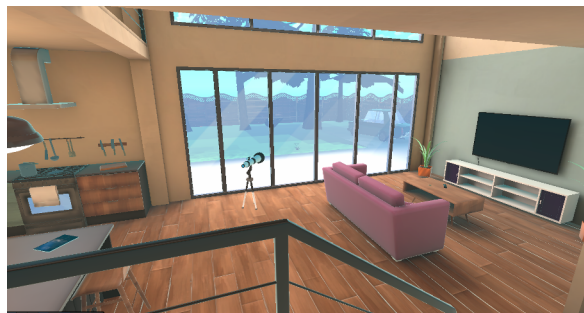


Figure 4: Living room environment. Screenshot from the ABA-VR application.



Figure 5: Supermarket environment. Screenshot from the ABA-VR application.

4.3 Interactions

Props such as fidget spinners, phones, tablets and TVs were created. These can be interacted with by both users. Objects can be taken

from the hand of another user, allowing the teacher to confiscate items and the student to grab items from the teacher. Additionally, in the supermarket environment, all items on the shelves are fully interactable and networked. Through the use of the contextual environment and interactive objects, the ABA expert is able to design scenarios for effective training that, to an extent, mimics the real world.

Furthermore, the child avatars are able to make facial expressions, seen in figure 6. The expressions are mapped to the VR controller buttons and synced across the network. This adds another layer of emotional response that the student is capable of using in certain contexts. Using a mixture of facial expressions, voice chat, synced hand poses and interactable objects. We believe the users are able to practice a wide range of practice scenarios.

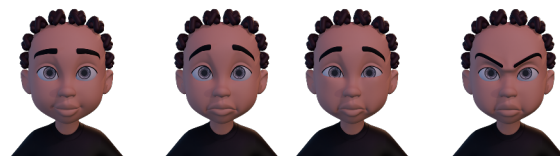


Figure 6: Examples of the available facial expressions. Renderings of avatars from the ABA-VR application.

5 CONCLUSION

This position paper highlights the critical need for accessible and effective training for educators working with students with ASD. Traditional ABA training methods face significant challenges, particularly in underfunded or geographically remote areas, where access to high-quality training is limited. The proposed VR platform addresses these challenges by offering immersive, cost-effective, and scalable training opportunities, bridging the gap between theoretical knowledge and practical application.

By leveraging VR technology, educators can engage in realistic, contextual scenarios, allowing them to develop critical ABA skills in a safe and controlled environment. The platform's ability to democratise access to training has the potential to reduce disparities in special education and improve classroom outcomes for both students and teachers.

Although this paper presents a promising solution for democratisation of ABA training, there are multiple avenues for improving the solution. Additional environments such as outdoor activities or public transportation could broaden the possible use cases in training. The practical impact of the solution remains speculative until it can be evaluated.

Future work will focus on empirically validating the platform through pilot studies and longitudinal research, as well as expanding its capabilities to address a broader range of behavioural challenges. Collaboration among educators, researchers, policymakers, and technology developers will be essential to ensure the successful adoption and scalability of VR-based ABA training.

We call on stakeholders across education and technology sectors to invest in innovative solutions like this platform to develop ecologically valid training simulations, equipping special education teachers with the skills to support children with ASD effectively.

ACKNOWLEDGMENTS

This project has been funded by the European Union ERASMUS+ program. The authors would like to thank our ABA partners at: Vi-Modo, Sina Svetulka, Masaryk University, SOSU Østjylland, Donbass State Pedagogical University and Blue Beehive.

REFERENCES

- [1] F. J. Alves, E. A. D. Carvalho, J. Aguilár, L. L. D. Brito, and G. S. Bastos. Applied behavior analysis for the treatment of autism: A systematic review of assistive technologies, 2020. doi: 10.1109/ACCESS.2020.3005296 [2](#)
- [2] J. Brede, A. Remington, L. Kenny, K. Warren, and E. Pellicano. Excluded from school: Autistic students' experiences of school exclusion and subsequent re-integration into school. *Autism & Developmental Language Impairments*, 2:2396941517737511, 2017. doi: 10.1177/2396941517737511 [2](#)
- [3] J. O. Cooper, T. E. Heron, and W. L. Heward. *Applied Behavior Analysis*. Pearson Education, 2007. [1](#), [2](#)
- [4] S. M. Edelson. Comparison of autistic individuals who engage in self-injurious behavior, aggression, and both behaviors. *Pediatric Reports*, 13(4):558–565, Oct. 2021. doi: 10.3390/pediatric13040066 [1](#)
- [5] L. Franz and G. Dawson. Implementing early intervention for autism spectrum disorder: A global perspective. *Pediatric Medicine*, 2(28), 2019. doi: 10.21037/pm.2019.07.02 [2](#)
- [6] I. M. Grey, R. Honan, B. McClean, and M. Daly. Evaluating the effectiveness of teacher training in applied behaviour analysis. *Journal of Intellectual Disabilities*, 9:209 – 227, 2005. [1](#)
- [7] C. A. Kearney. School absenteeism and school refusal behavior in youth: A contemporary review, 3 2008. doi: 10.1016/j.cpr.2007.07.012 [2](#)
- [8] N. Leonard and S. Smyth. Does training matter? exploring teachers' attitudes towards the inclusion of children with autism spectrum disorder in mainstream education in ireland. *International Journal of Inclusive Education*, 26:1–15, 01 2020. doi: 10.1080/13603116.2020.1718221 [2](#)
- [9] M. J. Maenner. Prevalence and characteristics of autism spectrum disorder among children aged 8 years—autism and developmental disabilities monitoring network, 11 sites, united states, 2018. *MMWR. Surveillance Summaries*, 70, 2021. [1](#)
- [10] J. L. Matson and S. V. LoVullo. Evidence-based practice for children with autism spectrum disorders. *Behavior Modification*, 32(5):583–607, 2008. doi: 10.1177/0145445507304581 [2](#)
- [11] C. Rice. Prevalence of autism spectrum disorders — autism and developmental disabilities monitoring network, six sites, united states, 2000, 2000. [1](#)
- [12] K. E. Stavroulia, I. Aedo, P. Díaz, and A. Lanitis. Virtual-reality based crisis management training for teachers: An overview of the vrteacher project. In *2021 International Symposium on Computers in Education (SIIE)*, pp. 1–4, 2021. doi: 10.1109/SIIE53363.2021.9583628 [2](#)
- [13] K. E. Stavroulia, A. Pyriní, I. Álvarez Testillano Siglo, S. S. Keegan, M. Debattista, T. Z. Ayo, C. Roussias, M. S. P. Siglo, P. D. Pérez, and A. Lanitis. Vrteacher project: A revolutionary approach to teacher education using virtual reality. [2](#)
- [14] O. I. Talantseva, R. S. Romanova, E. M. Shurdova, T. A. Dolgorukova, P. S. Sologub, O. S. Titova, D. F. Kleeva, and E. L. Grigorenko. The global prevalence of autism spectrum disorder: A three-level meta-analysis, 2 2023. doi: 10.3389/fpsy.2023.1071181 [1](#)
- [15] V. Totsika, R. P. Hastings, Y. Dutton, A. Worsley, G. Melvin, K. Gray, B. Tonge, and D. Heyne. Types and correlates of school non-attendance in students with autism spectrum disorders. *Autism*, 24:1639–1649, 10 2020. doi: 10.1177/1362361320916967 [2](#)